ECG ARRHYTHMIA BEAT CLASSIFICATION USING A CONVOLUTIONAL BIDIRECTIONAL LSTM DEPLOYED AS A TOOL FOR A RETRIEVAL-AUGMENTED LLM AGENT

A project report submitted in partial fulfilment of the requirements

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MASTER OF SCIENCE IN DATA SCIENCE

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BONAFIDE CERTIFICATE

This is to certify that the project report titled **"ECG Arrhythmia Beat Classification Using a Convolutional Bidirectional LSTM Deployed as a Tool for a Retrieval-Augmented LLM Agent"** is a bonafide record of the work carried out by **PRIYATOSH SATPATI** (Registration Number **25113170096**) in partial fulfilment of the requirements for the award of the degree of Master of Science in Data Science of SASTRA Deemed to be University, during the fourth semester of the programme, under my guidance and supervision.

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DECLARATION

I, **PRIYATOSH SATPATI** (Registration Number **25113170096**), declare that the project report titled **"ECG Arrhythmia Beat Classification Using a Convolutional Bidirectional LSTM Deployed as a Tool for a Retrieval-Augmented LLM Agent"** submitted to SASTRA Deemed to be University is a record of original work done by me under the guidance of Mr. G. Swaminathan. The source code, experiments, results and text are my own work. All datasets, publications and software used are acknowledged and cited in the References. The project was carried out at SASTRA Deemed to be University. This report does not contain any confidential information or the name of any customer, and it has not been submitted for the award of any other degree or diploma.

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PRIYATOSH SATPATI

ABSTRACT

Cardiac arrhythmias are usually identified from long ambulatory electrocardiogram (ECG) recordings, where manual beat-by-beat review is slow and error-prone. Many published deep-learning heartbeat classifiers report near-perfect accuracy, but they often split beats randomly, so beats from the same patient appear in both training and test data and the reported performance does not carry over to new patients. This project builds a heartbeat classifier that is evaluated on unseen patients and makes it usable through a conversational assistant.

Single-lead recordings from the MIT-BIH Arrhythmia Database are band-pass filtered, segmented around each R-peak and grouped into the AAMI classes normal (N), supraventricular ectopic (S) and ventricular ectopic (V). A compact convolutional bidirectional long short-term memory (Conv-BiLSTM) network of about 50,000 parameters combines beat morphology with rhythm features expressed relative to each patient's own heart rate. Model selection uses patients held out from training, and final testing uses 22 patients never seen during development, reported with patient-level bootstrap confidence intervals and compared against baselines.

On the unseen patients the model reaches 94.5% accuracy and a macro-F1 score of 0.65, against 0.53 for a rhythm-only baseline. It detects 80% of ventricular beats with 85% precision, and its probabilities are well calibrated. Supraventricular beats remain difficult, and this limitation is reported explicitly.

The model is deployed in a FastAPI web service with an ECG signal-quality check and is exposed as a callable tool to a large language model agent. The agent combines the classifier's output with retrieval-augmented generation over a persistent Chroma vector database of reference documents, so its answers about an uploaded ECG rest on both the measurement and cited sources. The system is verified by 95 automated tests covering preprocessing, APIs, integration, accuracy and performance.

Keywords: electrocardiogram, arrhythmia, heartbeat classification, Conv-BiLSTM, inter-patient evaluation, MIT-BIH, retrieval-augmented generation, LLM tool calling, Chroma, FastAPI.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AAMI	Association for the Advancement of Medical Instrumentation
API	Application Programming Interface
AUC	Area Under the ROC Curve
BiLSTM	Bidirectional Long Short-Term Memory
CI	Confidence Interval
CNN / Conv1D	(One-dimensional) Convolutional Neural Network
DS1 / DS2	Training / test record sets of de Chazal et al. [3]
ECE	Expected Calibration Error
ECG	Electrocardiogram
ERD	Entity Relationship Diagram
F / Q	Fusion / unclassifiable (paced) beats
LLM	Large Language Model
MLII	Modified limb lead II
N / S / V	Normal / supraventricular ectopic / ventricular ectopic beats
PPV	Positive Predictive Value (precision)
PVC	Premature Ventricular Contraction
RAG	Retrieval-Augmented Generation
ROC	Receiver Operating Characteristic
RR interval	Time between consecutive R-peaks
Se	Sensitivity (recall)
SDNN	Standard deviation of normal-to-normal RR intervals

CHAPTER 1: INTRODUCTION

1.1 Background

The electrocardiogram (ECG) records the electrical activity of the heart and is the primary tool for detecting arrhythmias, i.e. disturbances of the heart's rhythm. Many arrhythmias are intermittent, so patients are monitored with ambulatory (Holter) recorders for 24 hours or longer. A single day contains roughly 100,000 heartbeats, and reviewing them manually is slow, tiring and prone to error. Automatic heartbeat classification therefore has clear clinical value as a screening aid.

Two developments shape this project. First, deep learning models that combine convolutional and recurrent layers can learn beat morphology directly from the raw signal [8]–[11]. Second, large language models (LLMs) can call external tools and ground their answers in retrieved documents [14]–[16]. Combining the two allows a clinician or researcher to ask questions in natural language about an uploaded ECG and receive an answer based on an actual measurement and on cited reference material, rather than on the LLM's general knowledge.

1.2 Problem Statement

Published ECG classifiers frequently report accuracies of 98–99%, but many of them split heartbeats randomly into training and test sets. Because beats from one patient are highly similar, the model is then tested on people it has already seen, and the result does not describe its performance on a new patient [3], [13]. In addition, a trained classifier on its own is difficult for non-programmers to use and gives no context for its output.

The problem addressed is therefore: **to build an ECG heartbeat classifier whose performance is measured on patients it has never seen, to report that performance honestly with its uncertainty and limitations, and to make the classifier usable through a conversational agent whose answers are grounded in the model's output and in a searchable knowledge base.**

1.3 Objectives

1. Explore the MIT-BIH Arrhythmia Database and build a preprocessing pipeline (filtering, R-peak detection, beat segmentation, rhythm features) shared by training and deployment.
2. Train a compact Conv-BiLSTM classifier for the AAMI classes N, S and V that runs on an ordinary laptop CPU.
3. Evaluate it with a strict inter-patient protocol, baselines, patient-level confidence intervals, end-to-end detection, calibration and speed measurements.

4. Deploy the model in a FastAPI service with a persistent Chroma vector store for retrieval-augmented generation (RAG).
5. Expose the classifier and the knowledge base as tools to an LLM agent.
6. Verify the system with automated unit, API, integration, model-accuracy and model-performance tests.

1.4 Scope

The system classifies individual beats of a single-lead ECG into N, S and V. Fusion (F) and paced or unclassifiable (Q) beats are outside the trained classes; how the model labels them is reported. The system is a research and education tool and is **not** a medical device. The LLM is accessed through any OpenAI-compatible endpoint and is not part of the trained model.

1.5 Organisation of the Report

Chapter 2 reviews related work. Chapter 3 describes the dataset. Chapter 4 presents the methodology, Chapter 5 the system design including the entity relationship diagram, and Chapter 6 the implementation. Chapter 7 describes the testing strategy and test cases, and Chapter 8 the results. Chapter 9 discusses limitations and ethics, and Chapter 10 concludes.

CHAPTER 2: LITERATURE REVIEW

2.1 Data and Evaluation Standards

The MIT-BIH Arrhythmia Database [1], distributed through PhysioNet [2], is the most widely used benchmark for beat classification. The AAMI EC57 standard [4] groups its beat annotations into five classes (N, S, V, F, Q) and defines how sensitivity and positive predictivity are reported. De Chazal et al. [3] proposed splitting the 44 non-paced records into two sets of 22 patients (DS1 for training, DS2 for testing) so that no patient contributes to both. This *inter-patient* protocol has become the reference for realistic evaluation.

2.2 Feature-Based Methods

Early systems detected the QRS complex with the Pan–Tompkins algorithm [5] and classified beats from hand-crafted morphology and RR-interval features with linear discriminants [3] or ensembles of classifiers [12]. Llamedo and Martínez [13] showed that selecting features for generalisation across databases improves performance on unseen patients, and that RR-interval features normalised to the patient's own rhythm are among the most robust inputs.

2.3 Deep Learning Methods

Kiranyaz et al. [8] used one-dimensional CNNs on raw beats, Acharya et al. [9] trained a nine-layer CNN, and Yildirim [11] used bidirectional LSTMs [6], [7] on wavelet sequences. Hannun et al. [10] trained a deep network on a very large single-lead dataset and reported cardiologist-level performance on rhythm classes. Many deep-learning papers on MIT-BIH, however, use random beat-level splits; their very high accuracies are not comparable to inter-patient results, which are substantially lower, especially for supraventricular beats [3], [13].

2.4 Calibration and Uncertainty

Guo et al. [21] showed that modern neural networks are often over-confident and proposed the expected calibration error (ECE) to measure the gap between confidence and accuracy. The Brier score [24] measures the overall quality of probabilistic predictions. Because beats from one patient are correlated, confidence intervals should be computed by resampling patients rather than beats, using the bootstrap [20].

2.5 LLM Agents, Tool Use and RAG

Retrieval-augmented generation [14] grounds an LLM's answer in passages retrieved from an external store, reducing unsupported statements and allowing sources to be cited. ReAct [15] interleaves reasoning with tool calls, and Toolformer [16] showed that

language models can learn when to call external APIs. Dense retrieval relies on sentence embeddings such as Sentence-BERT [17]; MiniLM [18] provides a small embedding model that runs on a CPU.

2.6 Research Gap

Few works combine (i) a beat classifier evaluated with a strict inter-patient protocol and patient-level uncertainty, (ii) deployment as a service with tests that guard its accuracy, and (iii) integration as a tool for an LLM agent with retrieval over reference documents. This project addresses that combination with modest hardware and fully open data.

CHAPTER 3: DATASET AND DATA UNDERSTANDING

3.1 MIT-BIH Arrhythmia Database

The database contains 48 half-hour two-channel ambulatory ECG recordings from 47 subjects, digitised at 360 Hz with 11-bit resolution over a 10 mV range [1]. Every beat was annotated independently by two or more cardiologists. The database is available on PhysioNet [35]. This project uses the modified limb lead II (MLII), which is available in all non-paced records. The data is downloaded from PhysioNet [2] with the WFDB Python package and is distributed under the Open Data Commons Attribution License.

3.2 AAMI Class Mapping

Table 1: Mapping of MIT-BIH beat symbols to AAMI EC57 classes [4]

AAMI class	MIT-BIH symbols	Meaning
N	N, L, R, e, j	Normal, left/right bundle branch block, atrial/nodal escape
S	A, a, J, S	Atrial, aberrated atrial, nodal and supraventricular premature beats
V	V, E	Premature ventricular contraction, ventricular escape
F	F	Fusion of ventricular and normal beat (not modelled)
Q	/, f, Q	Paced, fusion of paced and normal, unclassifiable (not modelled)

3.3 Patient-Wise Data Split

The four paced records (102, 104, 107, 217) are excluded. The remaining records follow the inter-patient split of de Chazal et al. [3]. Five DS1 patients are held out as a validation set for early stopping and model selection, so the DS2 test patients are used only once, for the final evaluation (Table 2).

Table 2: Patient-wise split and beat counts per class

Set	Records	Patients	N	S	V	Purpose
Train	101, 108, 109, 112, 114, 115, 116, 119, 122, 203, 207, 208, 209, 215, 220, 223, 230	17	36,436	686	2,936	Fit weights
Validation	106, 118, 124, 201, 205	5	9,412	258	852	Early stopping, model selection
Test (DS2)	100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234	22	44,240	1,837	3,220	Final evaluation only

3.4 Observations from Exploratory Analysis

- **Severe class imbalance.** Normal beats make up about 90% of all beats, so a model that always predicts N already reaches about 90% accuracy. Accuracy alone is therefore misleading, and macro-F1 is used for model selection.

- **Patient concentration.** Rare classes are concentrated in few patients. In the test set, 1,382 of the 1,837 S beats come from a single record (232), and 373 of the 415 fusion beats available for training come from record 208.
- **Rhythm is informative.** Premature beats arrive early relative to the patient's own rhythm and are often followed by a compensatory pause, which motivates the patient-relative RR features of Section 4.1.
- **Noise and baseline wander.** The power spectrum shows low-frequency baseline wander and high-frequency noise, removed by a 0.5–40 Hz band-pass filter (notebook 01).

CHAPTER 4: METHODOLOGY

4.1 Preprocessing Pipeline

The same module (app/ecg/preprocessing.py) is used by the training notebooks and by the deployed service, so the beats seen at inference are built exactly like the training beats.

1. **Resampling** to 360 Hz, so recordings from devices sampling at, for example, 250 Hz are accepted.
2. **Band-pass filtering** at 0.5–40 Hz with a third-order zero-phase Butterworth filter.
3. **R-peak detection**: annotated peaks during training; at inference a Pan–Tompkins-style detector [5] (derivative, squaring, 150 ms moving-window integration, adaptive threshold, 250 ms refractory period, refinement to the local maximum) or peaks supplied by the caller.
4. **Segmentation**: a 256-sample window per beat, 90 samples (0.25 s) before and 166 samples (0.46 s) after the R-peak, z-score normalised per beat.
5. **Rhythm features**, all relative to the patient's own rhythm: previous and next RR interval divided by the local mean RR of about 10 beats, previous RR divided by the recording's median RR, and the ratio of previous to next RR. Absolute RR values in seconds are deliberately excluded, because they identify individual patients and caused over-fitting in an earlier version.
6. **Signal-quality index**: the excess kurtosis of each 10-second window of the filtered signal. QRS complexes make a clean ECG strongly peaked (median kurtosis about 10 on MIT-BIH), while noise and flat or periodic artefacts score near or below 0 [28]. A recording with more than half of its windows below 1.5 is flagged as unreliable. This check was added after an integration test showed that the detector reports beats in pure noise (Section 7.6).

4.2 Model Architecture

The classifier has two inputs (Figure 1). The beat waveform passes through two convolutional blocks that learn local morphology (QRS width, T-wave shape) and shorten the sequence from 256 to 64 steps, followed by two bidirectional LSTM layers [6], [7] that model the temporal order of the waves. The four rhythm features pass through batch normalisation [22] and a dense layer. Both branches are concatenated and classified by a dense layer with dropout [23] and a three-way softmax. Gaussian noise layers act as data augmentation during training only. The network has 49,571 parameters (0.71 MB).

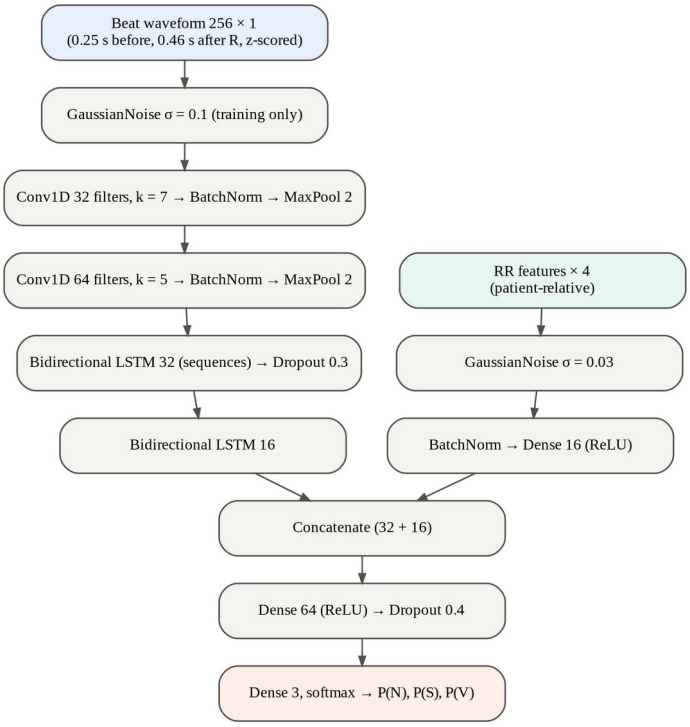


Figure 1: Conv-BiLSTM architecture with waveform and rhythm branches

4.3 Training Procedure

Table 3: Training configuration

Setting	Value	Reason
Optimiser	Adam [19], learning rate 0.001, halved on plateau	Standard, robust default
Batch size / max epochs	512 / 20	Fits a 2-core laptop CPU (about 60 s per epoch)
Loss	Categorical cross-entropy with balanced class weights	Counteracts the 90% majority of N beats
Model selection	Validation macro-F1, early stopping (patience 4)	Accuracy is dominated by N; loss does not reflect rare classes
Augmentation	Gaussian noise σ = 0.1 on beats, 0.03 on RR features	Robustness to noise and unseen morphology
Seed	42	Reproducibility

Training stopped after 8 epochs; the checkpoint from epoch 4 had the best validation macro-F1 (0.679) and was kept (Figure 2). Validation macro-F1 fluctuates between epochs because the validation set contains only five patients; selecting on it, rather than on training loss, avoids keeping a model that has started to over-fit.

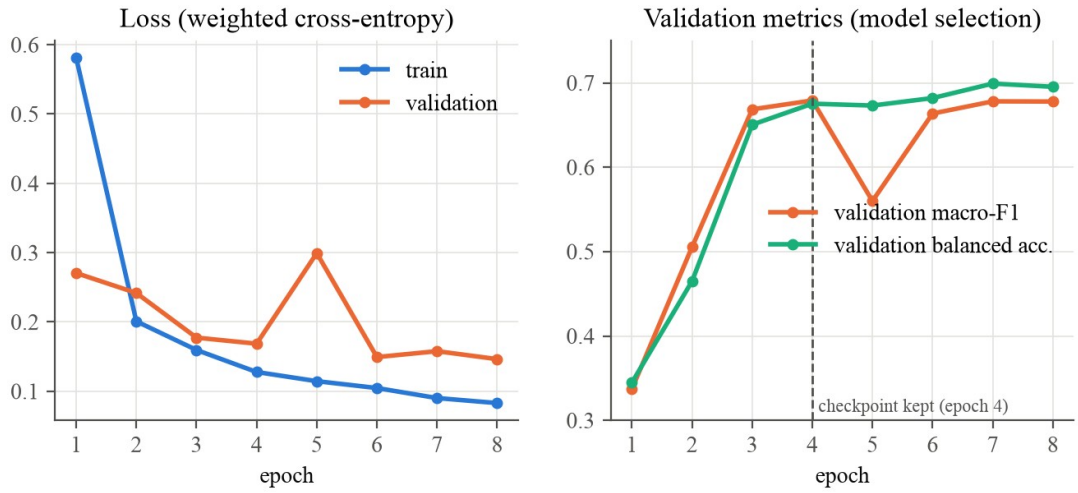


Figure 2: Training and validation curves; the dashed line marks the kept checkpoint

4.4 Evaluation Design

For each class c, with true positives TP, false negatives FN, false positives FP and true negatives TN:

$$\text{Sensitivity } Se = TP / (TP + FN) \qquad PPV = TP / (TP + FP) \qquad \text{Specificity} = TN / (TN + FP)$$

$$F1 = 2 \cdot Se \cdot PPV / (Se + PPV) \qquad \text{macro-}F1 = \text{mean of } F1 \text{ over N, S, V}$$

$$\text{balanced accuracy} = \text{mean of } Se \text{ over N, S, V}$$

- **Baselines**: a majority-class model (always N) and a logistic regression on the four RR features only, fitted on the same training patients.
- **Confidence intervals**: 1,000 bootstrap resamples of the 22 test *patients* [20]; beat-level resampling would understate the uncertainty.
- **End-to-end evaluation**: the deployed R-peak detector replaces the annotations; detected beats are matched to annotations within ±150 ms, as in AAMI EC57 [4].
- **Probability quality**: one-vs-rest ROC AUC, Brier score [24], ECE with 10 bins [21], and selective accuracy at confidence thresholds.
- **Significance**: a paired patient-level bootstrap comparing the macro-F1 of the model and the RR-only baseline on the same resampled patients.

4.5 Model Development History

Table 4: Model versions and why each was kept or rejected

Version	Change	DS2 macro-F1	Outcome
v1	4 classes, random beat-level validation split, absolute and relative RR features	not evaluated	Discarded: validation contained training patients (leakage)
v2	Patient-wise validation, macro-F1 selection, 4 classes (N/S/V/F)	0.458	Rejected: about 11% of normal beats labelled F (memorised patient 208); S Se 0.07

Version	Change	DS2 macro-F1	Outcome
v3	N/S/V only, patient-relative RR features, balanced weights, noise augmentation	0.650	Final model

CHAPTER 5: SYSTEM DESIGN

5.1 Architecture

Figure 3 shows the components. A FastAPI application exposes three groups of endpoints. Document endpoints extract and chunk text and store embedded chunks in a persistent Chroma collection. ECG endpoints store uploaded recordings and run the classifier. The chat endpoint runs an agent loop in which the LLM may call three tools: knowledge-base search, ECG analysis and listing of uploaded recordings. All services are created once at start-up (FastAPI lifespan) and shared by the requests.

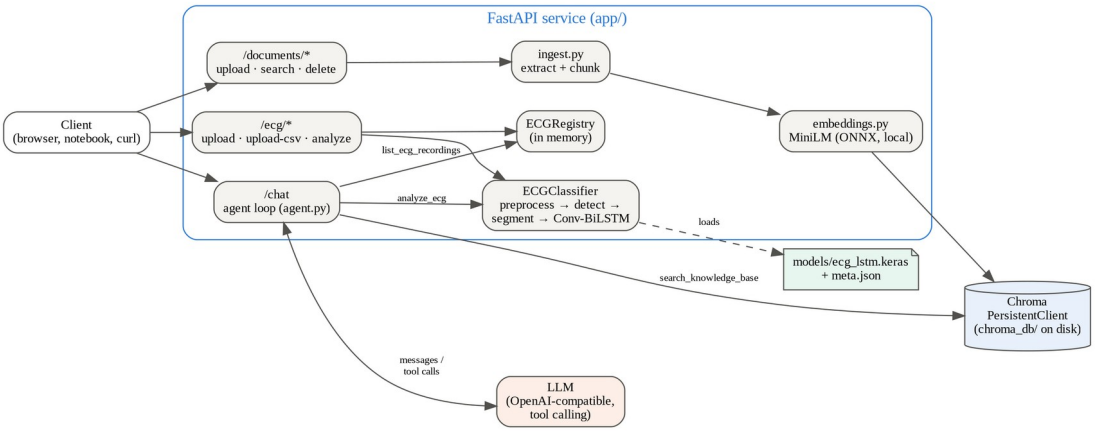


Figure 3: System architecture

5.2 Entity Relationship Diagram

Figure 4 shows the logical data model. Three groups of entities can be distinguished:

- **Training data (grey and green)**: a PATIENT_RECORD has many ANNOTATIONs and is segmented into many BEATs; beats train and evaluate a MODEL_VERSION, whose metadata (classes, hyperparameters, split, metrics) is stored in ecg_lstm_meta.json.
- **Runtime ECG analysis (blue)**: an ECG_UPLOAD is analysed by a MODEL_VERSION into an ANALYSIS_RESULT (heart rate, SDNN, beat counts, signal quality), which lists ABNORMAL_BEATs with time and confidence.
- **Knowledge base and agent (orange and pink)**: a Chroma COLLECTION stores DOCUMENTs, each split into CHUNKs with a 384-dimensional embedding and the identifier source::index. A CHAT_TURN may refer to an ECG_UPLOAD and makes TOOL_CALLs, which retrieve chunks or run an analysis.

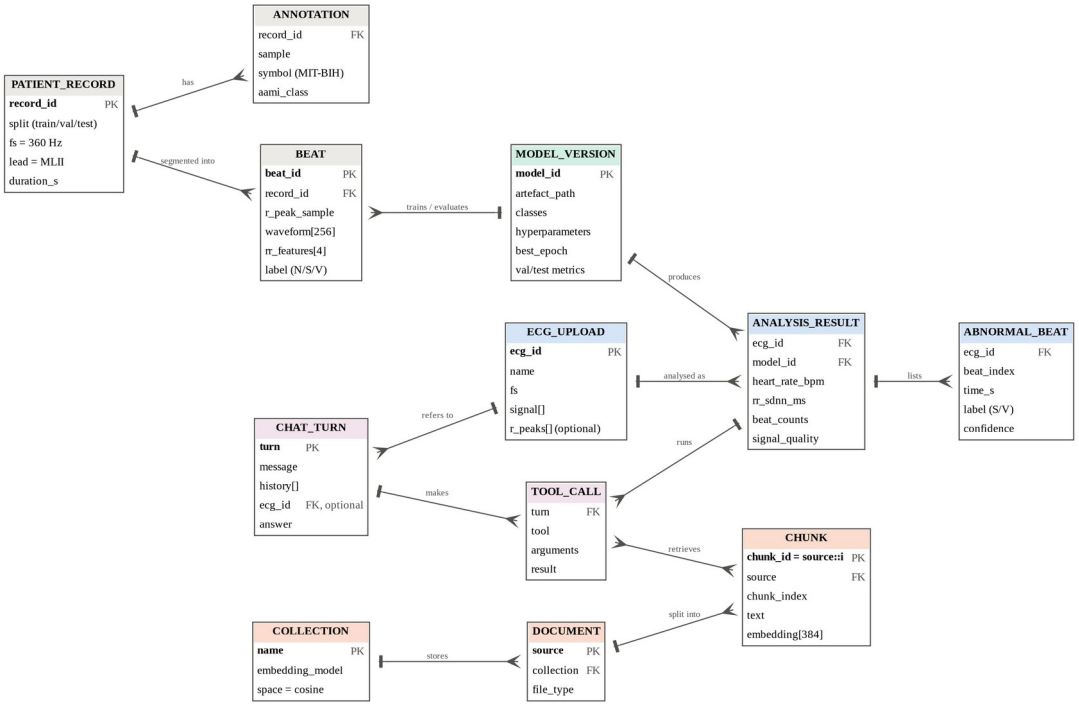


Figure 4: Entity relationship diagram (crow's foot notation)

Physically, chunks, embeddings and collection metadata live in Chroma's on-disk store (chroma_db/), the model and its metadata in models/, and uploaded ECGs in an in-memory registry. The collection records the name of the embedding model that built it, and the service refuses to mix embeddings from different models.

5.3 Request Flow of the Agent

Figure 5 shows a typical question about an uploaded ECG. The agent sends the conversation and the tool schemas to the LLM; the LLM requests tool calls; the agent executes them against the real classifier and vector store and returns the JSON results; the LLM then writes an answer grounded in those results. The loop is limited to six steps, and failures of the LLM provider are returned as HTTP 502 with a configuration hint.

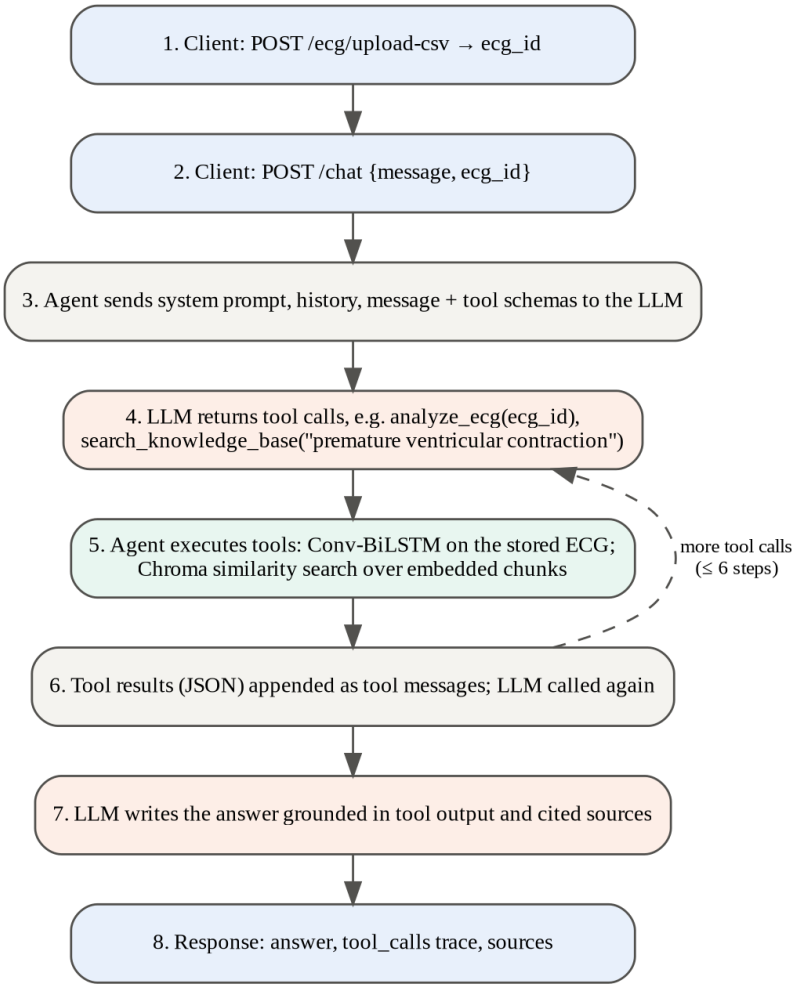


Figure 5: Request flow for a question about an uploaded ECG

5.4 API Endpoints

Table 5: REST endpoints

Method	Path	Purpose
GET	/health	Status, vector collection, whether the model is loaded
POST	/documents/upload	Upload PDF, TXT or Markdown; chunk, embed and store
POST	/documents/text	Ingest raw text under a source name
POST	/documents/search	Semantic search with optional source filter
DELETE	/documents/{source}	Remove all chunks of a document
GET	/documents/count	Number of stored chunks
POST	/ecg/upload	Upload an ECG as JSON (signal, sampling rate, optional R-peaks)
POST	/ecg/upload-csv	Upload an ECG as CSV
GET	/ecg/	List uploaded recordings
POST	/ecg/{ecg_id}/analyze	Run the classifier; returns heart rate, counts, abnormal beats, signal quality
POST	/chat	Agent with RAG and ECG tools

5.5 RAG Design

Documents are split with a recursive character splitter into chunks of 800 characters with 120 characters of overlap. Chunks are embedded locally with all-MiniLM-L6-v2 [17], [18] (ONNX runtime, no API key) or optionally with an OpenAI-compatible embedding endpoint. Chroma indexes the vectors with HNSW under cosine distance; search results are returned with a similarity score of 1 − distance. Re-uploading a document replaces its chunks instead of duplicating them.

CHAPTER 6: IMPLEMENTATION

6.1 Technology Stack

Table 6: Software used

Component	Library (version)
Language	Python 3.12
Deep learning	TensorFlow 2.21 / Keras 3.15 [26]
Signal processing, data	NumPy 2.5 [30], SciPy 1.18 [29], WFDB 4.3 [27], pandas 3.0
Classical ML and metrics	scikit-learn 1.9 [25]
Web service	FastAPI 0.139 [34], Uvicorn 0.51, Pydantic 2.13
Vector database	ChromaDB 1.5 (persistent client) [32]
Embeddings	all-MiniLM-L6-v2 via ONNX (Chroma default) [18]
LLM client	openai 2.45 (any OpenAI-compatible endpoint) [31]
Document parsing	pypdf 6.14, langchain-text-splitters 1.1
Testing	pytest 9.2, FastAPI TestClient, psutil
Notebooks and plots	Jupyter, Matplotlib 3.11
Hardware	Intel Core i5-7200U (2 cores, 4 threads), 16 GB RAM, no GPU

6.2 Project Structure

Table 7: Main modules and files

Path	Responsibility
app/ecg/preprocessing.py	Filtering, R-peak detection, segmentation, RR features, signal quality
app/ecg/mitbih.py	Download, loading and patient splits of MIT-BIH
app/ecg/classifier.py	Inference wrapper used by the API and the agent tool
app/ecg/evaluation.py	AAMI metrics, bootstrap CIs, end-to-end and probability metrics
app/vector_store.py, embeddings.py, ingest.py	Chroma store, embeddings, text extraction and chunking
app/agent.py	Tool schemas, system prompt, tool-calling loop
app/routers/*.py, main.py, config.py	REST endpoints, application start-up, settings from .env
notebooks/01–03	Exploratory analysis, training and validation, LSTM as an LLM tool
scripts/evaluate_model.py, benchmark_model.py	Accuracy report and performance benchmark (reports/)
scripts/test_report.py, make_figures.py	Test report and report figures
tests/	95 automated tests (Chapter 7)

6.3 Configuration and Notebooks

All settings (Chroma path and collection, LLM endpoint, key and model, embedding provider, chunking, model paths) are read from a .env file; a template (.env.example) is versioned while the real .env is excluded from version control. Notebook 01 explores

the data, notebook 02 trains, validates and tests the model and writes the model, its metadata and the training log, and notebook 03 demonstrates the model as an LLM tool, with and without RAG, both directly and through the REST API.

CHAPTER 7: TESTING AND VALIDATION

7.1 Strategy

Testing follows a layered approach. Fast unit and API tests use fakes for the LLM, the vector store and the classifier, so they run in seconds without a network or a trained model. Model, performance and integration tests run against the trained model and real MIT-BIH data, using only the 22 held-out DS2 patients. Thresholds are set a few points below the measured values, so the tests fail on real regressions rather than on normal variation between machines. The LLM is the only component that is always scripted, because it is an external paid service; the scripted LLM issues real tool calls that run against the real classifier and vector store.

Table 8: Test suites and results (95 of 95 passed)

Suite	Level	Scope	Tests	Passed
test_preprocessing.py	Unit	Signal processing, R-peak detection, RR features, signal quality	25	25
test_ingest.py	Unit	Text extraction (txt, md, pdf) and chunking	5	5
test_vector_store.py	Component	Real Chroma store on disk	5	5
test_agent.py	Unit	LLM tool-calling loop with a scripted LLM	8	8
test_api.py	API	Every endpoint with fakes; validation and error codes	14	14
test_model.py	Model	Correctness, robustness, accuracy floors on DS2	13	13
test_model_performance.py	Performance	Speed, memory, calibration, AUC, error rates, significance	14	14
test_api_integration.py	Integration	Real app: trained model, Chroma, local embeddings	11	11

7.2 Model Test Cases

Table 9: Selected model accuracy and robustness test cases

ID	Test case	Expected result	Status
TC-M01	Train, validation and test record lists are disjoint; test set equals DS2	No patient in two sets	Pass
TC-M02	API inference path vs training pipeline on record 233	≤ 2% of beat labels differ	Pass
TC-M03	Predictions repeated on the same beats	Identical; probabilities sum to 1	Pass
TC-M04	Signal scaled ×3 and offset +0.5	Beat counts unchanged (±1)	Pass
TC-M05	Record resampled to 250 Hz	Beats, V count and heart rate within 2%, 10%, 2 bpm	Pass
TC-M06	White noise added at 25 dB SNR	V count within 10%, heart rate within 2 bpm	Pass
TC-M07	Record 100 (sinus rhythm)	Heart rate 65–85 bpm, ≥ 90% N	Pass
TC-M08	Record 233 (frequent PVCs)	≥ 70% of annotated V beats reported	Pass
TC-M09	Accuracy floors on DS2	Acc ≥ 0.92, macro-F1 ≥ 0.60, Se N ≥ 0.97, Se V ≥ 0.75, PPV V ≥ 0.78	Pass

ID	Test case	Expected result	Status
TC-M10	Comparison with baselines	Macro-F1 above majority and RR-only baselines	Pass
TC-M11	End-to-end detection on DS2	Detection Se ≥ 0.97, PPV ≥ 0.99	Pass
TC-M12	All 22 DS2 records through signal-quality check	None flagged	Pass
TC-M13	Flat and 3-second inputs	No crash; valid response	Pass

7.3 Model Performance Test Cases

Table 10: Model performance test cases (measured value in brackets)

ID	Test case	Expected result	Status
TC-P01	Model size	< 100k parameters, < 2 MB (49,571; 0.71 MB)	Pass
TC-P02	Batched inference speed (2,048 beats)	< 5 ms per beat (0.447 ms)	Pass
TC-P03	Full 30-minute record through analyze()	< 30 s (1.31 s)	Pass
TC-P04	Batch of 40 vs 40 single predictions	Equal within 1e-4	Pass
TC-P05	Ten repeated analyses of a 5-minute strip	Memory growth < 200 MB (no leak)	Pass
TC-P06	Calibration on DS2	ECE < 0.05, Brier < 0.15 (0.0126; 0.0924)	Pass
TC-P07	ROC AUC per class	V ≥ 0.95, N ≥ 0.88, S ≥ 0.70 (0.9836; 0.9257; 0.7638)	Pass
TC-P08	Selective accuracy at rising confidence	Monotonic; ≥ 0.95 accuracy at ≥ 85% coverage	Pass
TC-P09	Normal beats predicted as V	< 1% (0.33%)	Pass
TC-P10	V specificity	≥ 0.98 (0.990)	Pass
TC-P11	S sensitivity without record 232	≥ 0.30 (0.382)	Pass
TC-P12	V sensitivity per patient	≥ 0.7 in ≥ 75% of patients; none < 0.3	Pass
TC-P13	Bootstrap CI of macro-F1 vs majority baseline	Lower bound > 0.315 (0.544)	Pass
TC-P14	Paired patient bootstrap vs RR-only baseline	Model better in ≥ 90% of resamples	Pass

7.4 API and Integration Test Cases

Table 11: Selected API and integration test cases

ID	Test case	Expected result	Status
TC-A01	GET /health	200, collection name, model loaded flag	Pass
TC-A02	Upload, search and delete a document	Chunks stored, relevant hit first, count back to 0	Pass
TC-A03	Re-ingest a document with the same name	Old chunks replaced, not duplicated	Pass
TC-A04	Unsupported file type or empty text	400	Pass
TC-A05	Signal shorter than 256 samples, fs = 0, top_k = 0	422	Pass
TC-A06	Analyse an unknown ecg_id	404	Pass
TC-A07	Analyse while the model is not loaded	503	Pass
TC-A08	Chat with the LLM provider unreachable	502 with a configuration hint	Pass
TC-A09	Chat history containing a system role	422 (prompt-injection guard)	Pass

ID	Test case	Expected result	Status
TC-I01	Start-up with real lifespan	Model loaded, Chroma collection created	Pass
TC-I02	Semantic search with real embeddings	ECG question → ECG document; baking question → cooking text	Pass
TC-I03	Restart the application	Knowledge base persists on disk	Pass
TC-I04	Record 233 uploaded as 250 Hz CSV	Beat count within 3%, V count within 15% of cardiologist labels	Pass
TC-I05	Upload with supplied R-peaks (record 100)	All peaks classified except edges; ≥ 95% N	Pass
TC-I06	Analyse response contract	All fields present; counts and percentages consistent	Pass
TC-I07	Upload and analyse a 5-minute strip over HTTP	< 5 s (measured ≈ 0.45 s)	Pass
TC-I08	Pure noise uploaded as an ECG	signal_quality.ok = false with a warning	Pass
TC-I09	Agent calls analyze_ecg and search on real services	Tool results contain the real model output and cited chunks	Pass

7.5 Mutation Testing

To check that the tests detect real faults, bugs were injected deliberately into the code [33] and the suite was re-run; every mutation was detected (Table below).

Table 12: Injected faults and the tests that detected them

Injected fault	Tests failed
RR features passed in the wrong order	Unit tests on RR features
Similarity score computed as distance instead of 1 − distance	Vector-store ranking tests
Tool result not returned to the LLM	Agent loop tests
API path ignores RR features	2 model tests
Signal-quality check disabled	4 tests (3 unit, 1 integration)
Rhythm features replaced by zeros at inference	7 of 14 performance tests (calibration, AUC, specificity, S detection, significance)

7.6 Defects Found by Testing

- **Patient leakage in validation (v1)**: a random beat split put training patients in the validation set. Fixed with patient-wise validation.
- **Over-prediction of fusion beats (v2)**: found by the held-out evaluation; fusion was removed from the trained classes.
- **Unhandled LLM errors**: an invalid key produced an HTTP 500; it now returns 502 with a hint.
- **Beats reported in pure noise**: an integration test showed that 20 s of Gaussian noise produced 39 "beats". A kurtosis-based signal-quality index was added; it flags all noise inputs and none of the 22 test recordings.

- **Edge-beat misalignment**: a consistency test exposed that segmentation drops edge beats; the training loader now records the R-peak of each kept beat.

CHAPTER 8: RESULTS AND DISCUSSION

8.1 Overall Performance on Unseen Patients

Table 13: Performance on the 22 DS2 test patients at annotated R-peaks (95% patient-bootstrap CI)

Model	Accuracy	Macro-F1	Balanced accuracy
Majority class (always N)	0.897	0.315	0.333
RR-only logistic regression	0.873	0.530	0.607
Conv-BiLSTM (this work)	0.945 [0.881–0.981]	0.650 [0.544–0.777]	0.630 [0.560–0.779]

Table 14: Per-class results on DS2

Class	Beats	Sensitivity [95% CI]	PPV [95% CI]	Specificity	F1
N	44,240	0.991 [0.984–0.997]	0.958 [0.904–0.990]	0.621	0.974
S	1,837	0.096 [0.006–0.498]	0.361 [0.069–0.514]	0.993	0.151
V	3,220	0.802 [0.604–0.926]	0.847 [0.543–0.970]	0.990	0.824

The model beats both baselines on macro-F1, and the lower bound of its confidence interval (0.544) is well above the majority baseline (Figure 6). Normal beats are recognised almost perfectly (Se 0.991), and ventricular beats are detected with sensitivity 0.802 and precision 0.847. Supraventricular beats are the main weakness, discussed in Section 8.5.

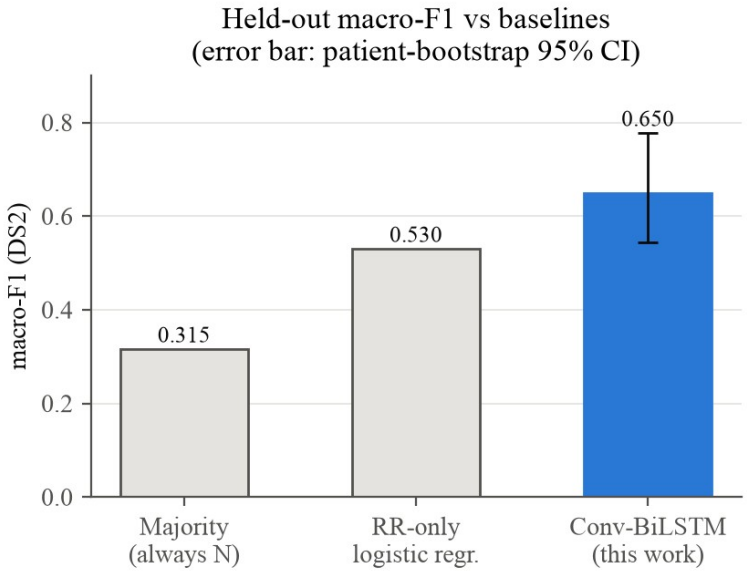


Figure 6: Held-out macro-F1 of the model and the baselines

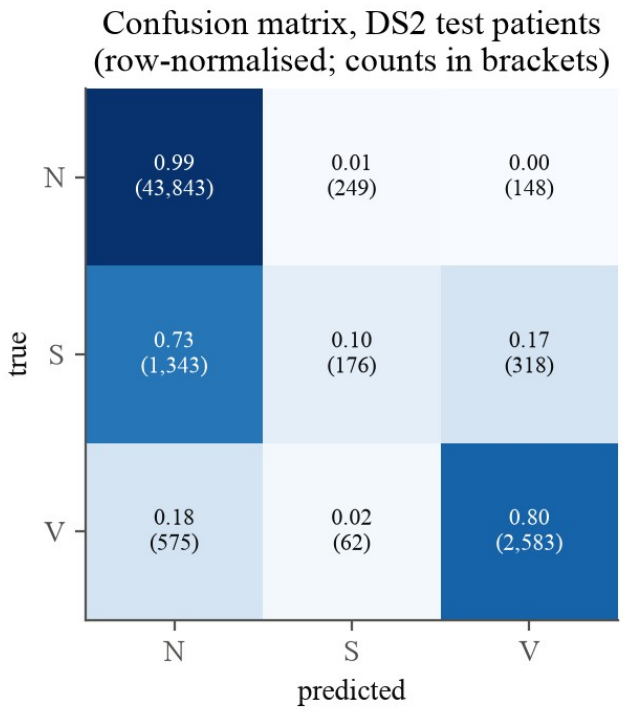


Figure 7: Confusion matrix on the DS2 test patients

8.2 End-to-End Performance

With the deployed R-peak detector instead of the annotations, 48,653 of 49,712 annotated beats are found (detection sensitivity 0.979, PPV 0.998, ±150 ms). Classification of the detected beats reaches accuracy 0.942 and macro-F1 0.642, with V sensitivity 0.805 and PPV 0.814. Performance therefore barely changes when the model runs on its own detected beats, which is the situation of a real user.

8.3 Calibration

The predicted probabilities are well calibrated: ECE is 0.0126, the Brier score 0.0924, and the reliability curve lies close to the diagonal (Figure 8). The ROC AUC is 0.9257 for N, 0.7638 for S and 0.9836 for V. Confidence is useful in practice: on the 91.5% of beats with confidence ≥ 0.9 the accuracy is 96.7%, so low-confidence beats can be highlighted for human review. The agent receives the confidence of every abnormal beat.

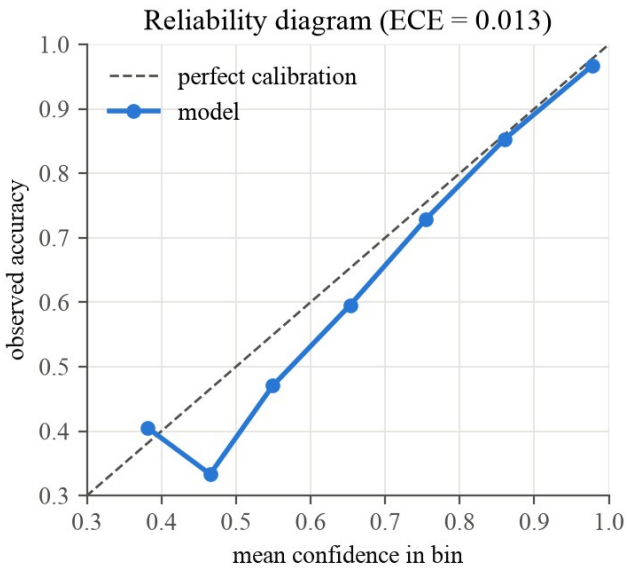


Figure 8: Reliability diagram of the predicted-class confidence on DS2

8.4 Computational Performance

Table 15: Speed and resources on an Intel i5-7200U laptop CPU without GPU

Measurement	Result
Inference, batch of 2,048 beats	0.447 ms per beat (2,237 beats/s)
Inference, single beat	78.01 ms (dominated by framework overhead)
Full analysis of a 30-minute record	1.31 s, about 1,373× faster than real time
API upload + analysis, 1-minute strip	160 ms (median of 5)
API upload + analysis, 5-minute strip	454 ms (median of 5)
Model size	49,571 parameters, 0.71 MB
Training time	8 epochs, about 9 minutes

8.5 Error Analysis

S beats are mostly missed (Se 0.096). 1,382 of the 1,837 S beats in DS2 come from record 232, a patient with sinus bradycardia and long pauses, in which atrial premature beats are not early relative to the patient's own rhythm, the main cue the model relies on. Without this record, accuracy is 0.972, macro-F1 0.740, and S sensitivity 0.382. The patient-level confidence interval for S sensitivity ([0.006–0.498]) reflects this dependence on individual patients.

V sensitivity also varies between patients: per record it ranges from 0.42 to 1.00. Only 0.33% of normal beats are wrongly reported as V, so false ventricular alarms are rare. Of the 388 fusion beats in DS2, 380 are labelled N, as expected for a class that was not trained.

Figure 9 shows ten seconds of record 233, a held-out patient. The model flags every ventricular beat with confidence above 0.95 and leaves the normal beats unflagged; the single S beat at 20.8 s is missed, illustrating the weakness described above.

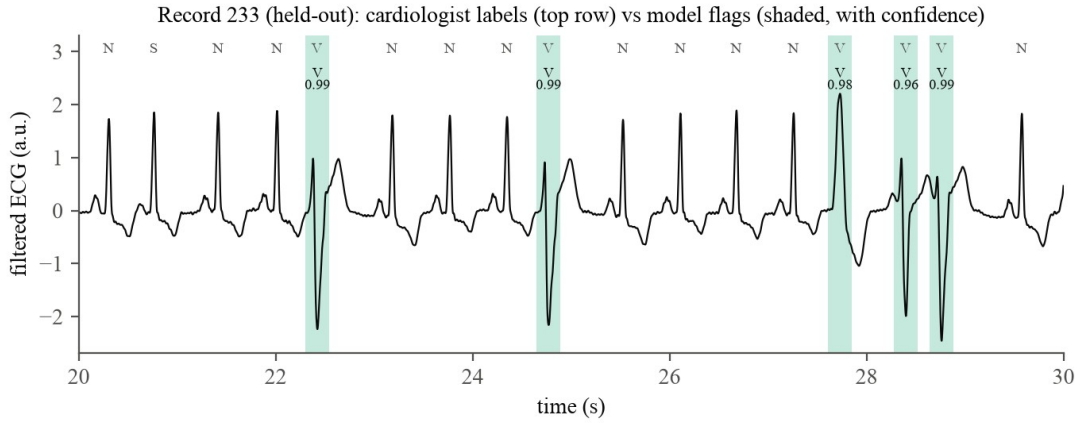


Figure 9: Model output on record 233 compared with cardiologist labels

8.6 Discussion

The results show that a small model trained on a laptop can reach useful performance on unseen patients for normal and ventricular beats, the most frequent and clinically most relevant ectopic class. They also confirm what the literature reports for the inter-patient setting [3], [13]: supraventricular beats are much harder to recognise on new patients than in intra-patient experiments. Reporting patient-level confidence intervals, baselines and a per-record breakdown makes this visible, where a single accuracy figure (94.5%) would hide it.

Integrating the model as an LLM tool keeps the measurement and the explanation separate. The classifier produces the numbers, the knowledge base supplies the reference text, and the LLM only combines them, with instructions to state the model's limitations, to mention its confidence and to disregard results flagged by the signal-quality check.

CHAPTER 9: LIMITATIONS AND ETHICAL CONSIDERATIONS

9.1 Limitations

- Low sensitivity for supraventricular beats; the model must not be used to rule out atrial ectopy.
- Fusion and paced beats are not recognised; recordings from patients with pacemakers are out of scope.
- Trained on a single lead (MLII) of a single database of 47 subjects recorded in 1975–1979; other leads, wearable devices and populations are untested.
- RR features use the recording's own rhythm as reference, so strips shorter than about 10 beats give less reliable results.
- Uploaded ECGs are kept in memory and are lost on restart; the service has no authentication.
- The quality of the agent's answers depends on the external LLM, which was exercised with a scripted LLM in the tests.

9.2 Ethical Considerations

- **Not a medical device.** Every analysis includes a disclaimer, and the agent is instructed to state that results are not a diagnosis.
- **Data privacy.** The MIT-BIH recordings are de-identified and openly licensed. ECGs uploaded by users are not written to disk.
- **Transparency.** Limitations, failure cases and rejected model versions are documented in the model card and in this report.
- **Confidentiality.** The project uses only public data and open-source software and contains no confidential or customer information.

CHAPTER 10: CONCLUSION AND FUTURE WORK

10.1 Conclusion

This project delivered a complete, tested system for ECG beat classification and its use by an LLM agent. A Conv-BiLSTM of 49,571 parameters, trained on a laptop CPU, reaches 94.5% accuracy and a macro-F1 of 0.650 on 22 unseen patients, beating a rhythm-only baseline in a paired patient-level bootstrap, with well-calibrated probabilities and an end-to-end pipeline that analyses a 30-minute recording in about a second. The model is served by a FastAPI application with a persistent Chroma knowledge base and is exposed as a tool to an LLM agent. 95 automated tests, checked by mutation testing, guard preprocessing, APIs, integration, accuracy and performance.

10.2 Contributions

1. A shared preprocessing pipeline with patient-relative rhythm features and a kurtosis-based signal-quality check.
2. A strict inter-patient evaluation with baselines, patient-bootstrap confidence intervals, end-to-end detection, calibration and error analysis.
3. A deployable service that exposes the model and a RAG knowledge base as tools to an LLM agent.
4. A test suite that includes accuracy floors, performance tests and integration tests on held-out patients.

10.3 Future Work

1. Improve S-beat detection with longer rhythm context (sequence-of-beats models) and patient-adaptive fine-tuning on a few labelled beats.
2. Train and validate on additional databases (e.g. INCART, the European ST-T database) and on wearable single-lead devices.
3. Add fusion and paced classes with more varied training data.
4. Add persistent storage, authentication and audit logging for the uploaded recordings.
5. Evaluate the agent's answers with clinicians and with automatic groundedness metrics.

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APPENDIX A: HOW TO REPRODUCE

All commands are run from the project root on Linux with Python 3.12.

```
python3 -m venv env && source env/bin/activate && pip install -r
requirements.txt
cp .env.example .env    # set LLM_API_KEY (and LLM_BASE_URL / LLM_MODEL
if not OpenAI)
jupyter nbconvert --to notebook --execute notebooks/01_ecg_analysis.ipynb
jupyter nbconvert --to notebook --execute
notebooks/02_lstm_training.ipynb
python scripts/evaluate_model.py      # reports/evaluation.md
python scripts/benchmark_model.py     # reports/performance.md
python scripts/test_report.py         # runs all tests,
reports/test_report.md
python scripts/make_figures.py        # docs/figures/
uvicorn app.main:app                  # API docs at
http://localhost:8000/docs
```

Model artefacts (models/), MIT-BIH data (data/) and the vector store (chroma_db/) are generated by these steps and are not stored in version control.

APPENDIX B: COMPLETE TEST CASE CATALOGUE

Generated from reports/junit.xml: 95 passed, 0 failed or skipped, of 95 tests.

Table 16: All automated test cases

#	Suite	Test case	Time (s)	Result
1	agent	tool schemas are valid	0.00	Pass
2	agent	registry roundtrip	0.00	Pass
3	agent	answer without tools	0.08	Pass
4	agent	agent runs ecg tool and feeds result back	0.03	Pass
5	agent	agent rag tool collects sources	0.03	Pass
6	agent	agent handles unknown ecg and bad arguments	0.03	Pass
7	agent	agent without model reports error	0.03	Pass
8	agent	agent stops after max steps	0.03	Pass
9	api	health	0.07	Pass
10	api	openapi lists all routes	0.07	Pass
11	api	upload search delete document	0.04	Pass
12	api	reingest replaces previous chunks	0.04	Pass
13	api	upload rejects unsupported and empty	0.04	Pass
14	api	search validates top k	0.03	Pass
15	api	ecg upload json and analyze	0.04	Pass
16	api	ecg upload csv skips header	0.04	Pass
17	api	ecg upload validation	0.04	Pass
18	api	ecg analyze unknown id	0.04	Pass
19	api	ecg analyze without model returns 503	0.04	Pass
20	api	chat end to end with tool call	0.04	Pass
21	api	chat llm failure returns 502 with hint	0.04	Pass
22	api	chat rejects unknown ecg and bad history	0.04	Pass
23	api_integration	startup loads real model and store	3.81	Pass
24	api_integration	ingest reference document and retrieve relevant chunks	0.98	Pass
25	api_integration	semantic search prefers matching document	0.87	Pass
26	api_integration	knowledge base persists across restart	0.66	Pass
27	api_integration	csv upload at 250hz matches cardiologist labels	1.42	Pass
28	api_integration	json upload with supplied r peaks	0.19	Pass
29	api_integration	analyze response contract	0.16	Pass
30	api_integration	analyze latency	0.97	Pass
31	api_integration	noise only input is flagged as unreliable	0.11	Pass
32	api_integration	real ecg passes signal quality	0.21	Pass
33	api_integration	agent uses real model and knowledge base	0.50	Pass
34	ingest	extract text txt and md	0.00	Pass
35	ingest	extract text pdf	0.00	Pass
36	ingest	extract text rejects unsupported	0.00	Pass
37	ingest	chunk text respects size and overlap	0.00	Pass
38	ingest	chunk text drops blank input	0.00	Pass
39	model	meta matches preprocessing and has no patient leakage	0.00	Pass
40	model	predictions are deterministic probabilities	1.97	Pass
41	model	api pipeline matches training pipeline	0.84	Pass
42	model	regular rhythm record	0.51	Pass
43	model	detects ventricular ectopy	0.43	Pass
44	model	invariant to gain and dc offset	0.40	Pass

#	Suite	Test case	Time (s)	Result
45	model	robust to lower sampling rate	0.71	Pass
46	model	robust to moderate noise	0.72	Pass
47	model	all test patients pass signal quality	4.32	Pass
48	model	short and flat inputs do not crash	0.12	Pass
49	model	ds2 accuracy floors	37.60	Pass
50	model	beats majority and rr only baselines	0.00	Pass
51	model	end to end detection and classification	39.20	Pass
52	model_performance	model is small enough for cpu deployment	0.00	Pass
53	model_performance	batched inference throughput	1.23	Pass
54	model_performance	full record analysed much faster than real time	1.46	Pass
55	model_performance	predictions do not depend on batch size	3.71	Pass
56	model_performance	memory stays bounded on repeated analysis	4.64	Pass
57	model_performance	probabilities are well calibrated	37.75	Pass
58	model_performance	roc auc per class	0.00	Pass
59	model_performance	higher confidence means higher accuracy	0.00	Pass
60	model_performance	few normal beats raised as ventricular	0.00	Pass
61	model_performance	v specificity	0.00	Pass
62	model_performance	s detection outside record 232	0.00	Pass
63	model_performance	v detected in most patients	0.01	Pass
64	model_performance	macro f1 ci lower bound beats baselines	0.00	Pass
65	model_performance	paired bootstrap vs rr baseline	6.32	Pass
66	preprocessing	beat window constants	0.00	Pass
67	preprocessing	aami mapping covers all classes	0.00	Pass
68	preprocessing	resample length[128]	0.00	Pass
69	preprocessing	resample length[250]	0.00	Pass
70	preprocessing	resample length[500]	0.00	Pass
71	preprocessing	resample length[1000]	0.00	Pass
72	preprocessing	resample noop at target fs	0.00	Pass
73	preprocessing	bandpass removes baseline wander keeps qrs band	0.01	Pass
74	preprocessing	normalize beat	0.00	Pass
75	preprocessing	normalize flat beat is finite	0.00	Pass
76	preprocessing	detect r peaks on synthetic ecg[0.5]	0.06	Pass
77	preprocessing	detect r peaks on synthetic ecg[0.8]	0.03	Pass
78	preprocessing	detect r peaks on synthetic ecg[1.2]	0.03	Pass
79	preprocessing	detect r peaks after resampling from 250hz	0.02	Pass
80	preprocessing	rr features regular rhythm	0.00	Pass
81	preprocessing	rr features are heart rate independent	0.00	Pass
82	preprocessing	rr features flag premature beat	0.00	Pass
83	preprocessing	rr features degenerate inputs	0.00	Pass
84	preprocessing	segment beats skips edges	0.00	Pass
85	preprocessing	segment beats empty	0.00	Pass
86	preprocessing	signal quality accepts synthetic ecg	0.03	Pass
87	preprocessing	signal quality rejects non ecg[gaussian]	0.01	Pass
88	preprocessing	signal quality rejects non ecg[sine]	0.00	Pass
89	preprocessing	signal quality rejects non ecg[flat]	0.00	Pass
90	preprocessing	signal quality is scale invariant	0.04	Pass
91	vector_store	add search ranks relevant chunk first	0.31	Pass
92	vector_store	data persists across restarts	0.30	Pass
93	vector_store	reingest upserts and source filter and delete	0.45	Pass

#	Suite	Test case	Time (s)	Result
94	vector_store	search empty collection and large top k	0.29	Pass
95	vector_store	refuses mismatched embedding model	0.30	Pass